

Carrying cost: from the annual rate r to the unit holding cost $h = r \cdot v$

Part 6 · Show me the money — solution report with data, method, results and charts

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1. Part 6 · Show me the money — Carrying cost: from the annual rate r to the unit holding cost $h = r \cdot v$

1. Carrying cost: from the annual rate r to the unit holding cost $h = r \cdot v$

$$r = 28\% + 2\% + 6\% + 1\% = 37\% \quad h = 0.37 \times 180 = 66.60 \quad \text{Total} = 5 \times 300 \times 66.60 = 99,900.00$$

37 % Annual carrying rate r	37 Piasters per EGP per year	\$66.60 /unit/yr h for the \$180 item
\$99,900 300 units $\times$ 5 years	76% Capital cost share of r	

Carrying cost — $r = 37\%$ per year (slide 45)

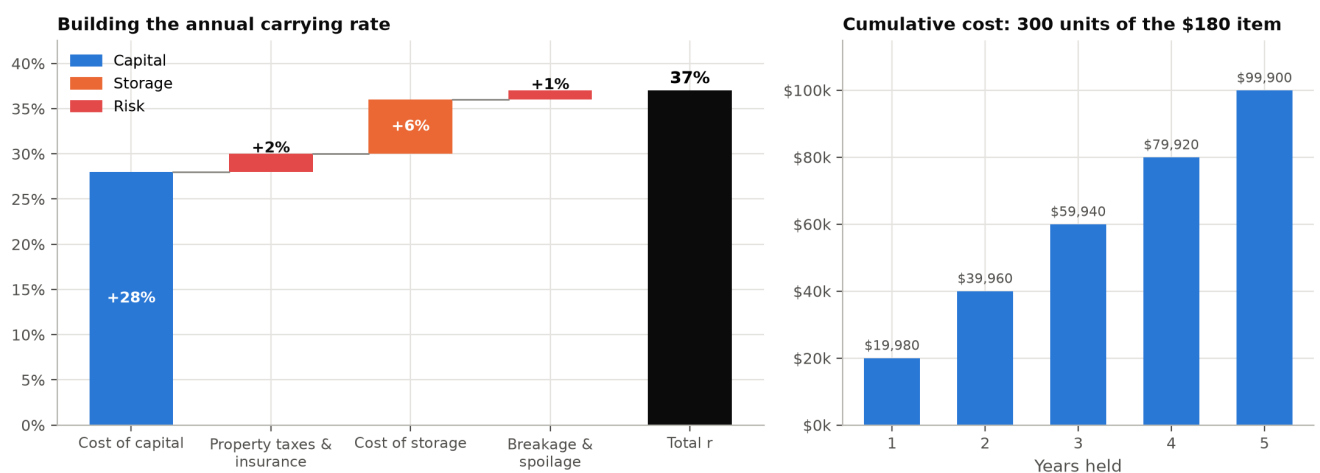


Figure 1.1 — Left: components of r (waterfall). Right: cumulative holding cost of the lecture example.

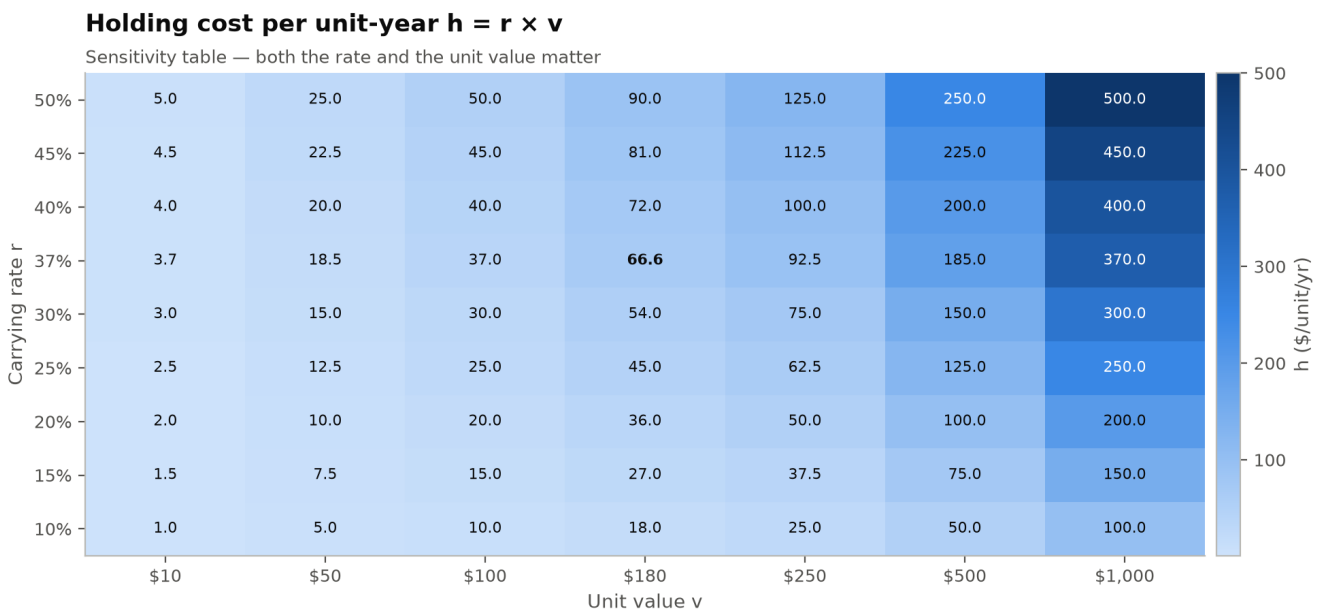
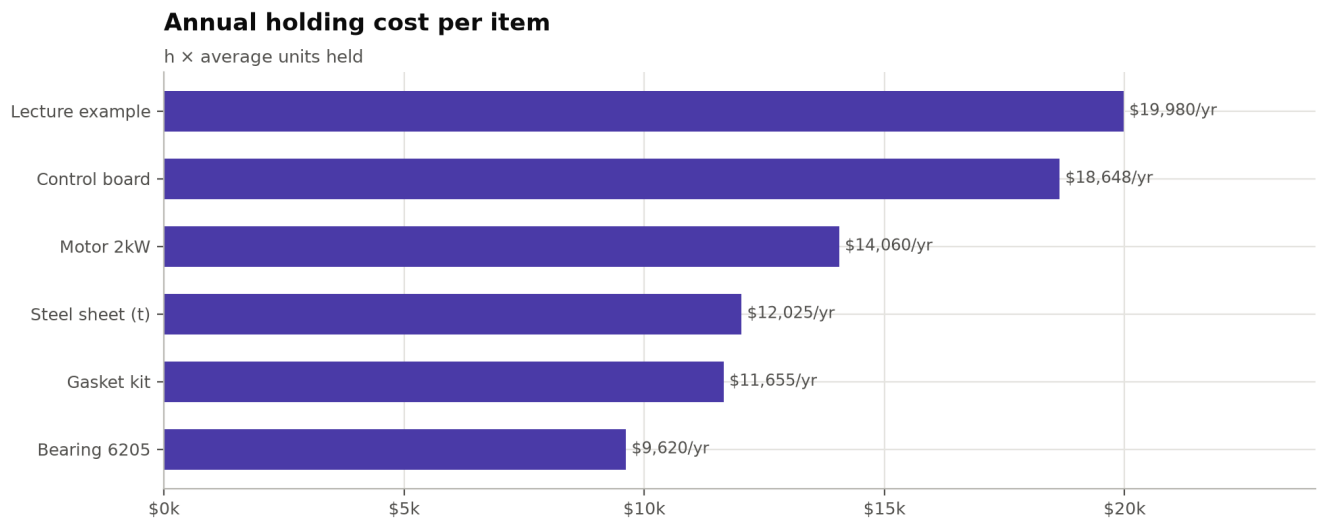


Figure 1.2 — h for combinations of r and v .

Figure 1.3 — Annual holding cost by item at $r = 37\%$.

Carrying-rate components.

component	group	annual rate pct
Cost of capital	Capital cost	28
Property taxes & insurance	Risk cost	2
Cost of storage	Storage cost	6
Breakage & spoilage	Risk cost	1

Holding cost by item.

item	unit value	avg units held	years held	h per unit year	holding cost per year	total holding cost
Lecture example	\$180.00	300	5	\$66.60	\$19,980	\$99,900
Motor 2kW	\$950.00	40	1	\$351.50	\$14,060	\$14,060
Control board	\$420.00	120	1	\$155.40	\$18,648	\$18,648
Steel sheet (t)	\$1,300.00	25	1	\$481.00	\$12,025	\$12,025
Gasket kit	\$35.00	900	1	\$12.95	\$11,655	\$11,655
Bearing 6205	\$6.50	4,000	1	\$2.40	\$9,620	\$9,620